

# Customer Churn Prediction & Retention Analytics

## 1. Introduction

Customer churn is one of the biggest challenges faced by telecom companies, as losing existing customers directly impacts revenue and profitability. Understanding the factors that influence customer churn enables businesses to develop effective retention strategies and improve customer satisfaction.

This project focuses on analyzing telecom customer data to identify churn patterns, predict customers who are likely to leave, and provide actionable business recommendations. The project integrates **SQL for data analysis, Python for machine learning, and Power BI for interactive dashboard development**, creating an end-to-end analytics solution for customer retention.

## 2. Abstract

The objective of this project is to analyze customer behavior and predict churn using historical telecom customer data. The dataset was first imported into MySQL for exploratory data analysis, where key performance indicators (KPIs) and churn trends were identified. Python was then used for data preprocessing, feature engineering, machine learning model development, and customer risk prediction. Finally, the processed dataset was visualized in Power BI through interactive dashboards that present executive insights, churn analysis, and customer risk prediction.

The developed solution helps identify high-risk customers, understand the major reasons behind customer churn, and recommend appropriate retention strategies for improving customer loyalty and reducing churn rates.

## 3. Tools Used

- **MySQL Workbench** – Database creation, SQL queries, and exploratory data analysis
- **Python (Google Colab)** – Data preprocessing, feature engineering, and machine learning
- **Power BI Desktop** – Interactive dashboard development and business visualization
- **Microsoft Excel** – Initial dataset cleaning and handling missing values

### Python Libraries

- Pandas
- NumPy
- Matplotlib
- Scikit-learn

## Machine Learning Models

- Logistic Regression
- Random Forest Classifier

## 4. Steps Involved in Building the Project

### Step 1: Data Collection & Preparation

- Imported the telecom customer churn dataset.
- Cleaned missing values and corrected data inconsistencies.
- Loaded the cleaned dataset into MySQL.

### Step 2: SQL-Based Exploratory Data Analysis

- Created the telecom customer database and table.
- Calculated key metrics such as total customers, churn rate, and churned customers.
- Analyzed churn based on contract type, payment method, internet service, and churn reasons.

### Step 3: Machine Learning

- Performed data preprocessing using label encoding and one-hot encoding.
- Split the dataset into training and testing sets.
- Built and evaluated Logistic Regression and Random Forest models.
- Generated churn probability and customer risk levels (Low, Medium, High).

### Step 4: Power BI Dashboard Development

Developed a three-page interactive dashboard consisting of:

- **Executive Dashboard** – Overall KPIs and churn overview.
- **Customer Churn Analysis** – Root cause analysis and business insights.
- **Risk Prediction Dashboard** – High-risk customer identification and retention strategies.

## 5. Conclusion

The project successfully demonstrates an end-to-end customer churn analytics pipeline by integrating SQL, Python, Machine Learning, and Power BI. The analysis identified major churn drivers such as month-to-month contracts, higher monthly charges, lack of technical support, and electronic check payment methods. The predictive model enables businesses to proactively identify high-risk customers and implement targeted retention strategies.

Overall, this project highlights how data analytics and machine learning can support data-driven decision-making, improve customer retention, and enhance long-term business performance.